



## E-Commerce Web Application

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### Track: Frontend Development

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# 1. Project Idea

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Volt Market is a fully responsive e-commerce web application that allows users to browse products, manage their shopping cart, and complete a smooth checkout experience — all powered by a real REST API and built with modern frontend technologies. The platform supports user authentication, protected routes, and real-time notifications.

# 2. Problem the Project Solves

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Many users struggle with e-commerce platforms that are slow, hard to navigate, or not optimized for mobile. Volt Market addresses this by delivering a fast, clean, and mobile-friendly shopping experience that is easy to use and visually appealing.

# 3. Tools & Technologies Used

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| Technology       | Version | Purpose                    |
|------------------|---------|----------------------------|
| React            | 19      | UI Library                 |
| React Router DOM | v7      | Client-side Routing        |
| Bootstrap        | 5       | Responsive Styling         |
| Axios            | 1.x     | API Requests               |
| Formik + Yup     | Latest  | Form Handling & Validation |
| React Hot Toast  | 2.x     | Notifications              |
| Vite             | 8       | Build Tool                 |
| GitHub Pages     | —       | Deployment                 |

# 4. Project Pages & Functions

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## 4.1 Home Page

The main landing page displaying featured products, categories, and promotional banners. Users can browse all available products and navigate to product details.

## 4.2 Products Page

Displays all products fetched from the API. Users can filter by category or brand and add items directly to their cart.

## 4.3 Product Details Page

Shows full details of a single product including images, price, description, and an add-to-cart button.

## 4.4 Cart Page

Shows all items the user has added to their cart with options to update quantities or remove items. Displays the total price and proceeds to checkout.

## 4.5 Brands Page

Lists all available brands. Clicking a brand filters products by that brand.

## 4.6 Login / Register Pages

Authentication pages with form validation using Formik and Yup. Successful login stores the token and redirects the user.

# 5. Core Features

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- User Authentication (Register / Login / Logout)
- Protected Routes for authenticated users only
- Browse and filter products by category and brand
- Add to cart and manage cart items
- Full checkout flow
- Real-time toast notifications
- Fully responsive on all screen sizes
- Fast performance with Vite build tool

# 6. Screenshots

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Screenshots of all pages are available in the /screenshots folder of the GitHub repository.

Live Demo: <https://anwarabotaleb-full-dev.github.io/Volt-Market/>

## 7. Challenges & Solutions

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| Challenge                                 | Solution                                                   |
|-------------------------------------------|------------------------------------------------------------|
| Managing state across multiple components | Used React Context API for global state management         |
| Implementing Protected Routes             | Created a ProtectedRoutes wrapper component                |
| API loading and error handling            | Used Axios interceptors and loading states                 |
| Responsive design across all screens      | Leveraged Bootstrap grid and custom CSS                    |
| Time management among team members        | Divided tasks by feature and used GitHub for collaboration |

## 8. Future Improvements

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- Add multi-language support (Arabic / English)
- Build an Admin Dashboard for product and order management

## 9. Conclusion

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Volt Market is a complete e-commerce frontend project that demonstrates the ability to build a real-world application using modern technologies. The team successfully delivered a responsive, API-integrated shopping platform with authentication and a clean user experience. The project is a strong foundation for future enhancements and real-world deployment.

## 10. Team Members

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| # | Name | Track |
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|---|---------------------------|----------------------|
| 1 | Anwar Khaled Abotaleb     | Frontend Development |
| 2 | Ahmed AbdElghafar Hammad  | Frontend Development |
| 3 | Hussein Khaled Elsharkawy | Frontend Development |
| 4 | Mahmoud Abdallah Arnous   | Frontend Development |
| 5 | Habiba Ramadan Elsayed    | Frontend Development |